Waves and sound

Subject: Physics | Level: Class 10 | Curriculum: Federal Board
Prepared by: SHS Virtual Academy

1. Topic Overview and Learning Objectives

Waves and sound form a fundamental chapter in the Class 10 Physics curriculum, serving as the gateway to understanding how energy travels through space and matter. In our daily lives, we are surrounded by waves, from the light that allows us to see to the sound waves that enable us to communicate. Mastering this topic is essential for grasping more advanced concepts in higher physics and engineering.

For the Federal Board examinations, this chapter is highly significant as it consistently features both conceptual questions and numerical problems. Examiners often test your ability to relate wave speed, frequency, and wavelength, as well as your understanding of how sound properties like pitch and loudness are influenced by physical characteristics. A strong grasp of these principles will secure you valuable marks in the long-answer section.

The fundamental intuition behind this topic is the concept of energy transfer without the transfer of matter. Whether it is a mechanical wave moving through a medium or sound vibrating through air, the particles only oscillate about their mean positions while the disturbance propagates forward. By visualizing these oscillations, you will find it much easier to derive formulas and solve the complex problems presented in your board exams.

2. Core Concepts and Detailed Revision Notes

2.1 Simple Harmonic Motion and Waves

Simple Harmonic Motion is the foundation upon which we build our understanding of waves. It occurs when a restoring force is directly proportional to the displacement from the mean position and is directed towards it. When this oscillatory motion is transferred through a medium, we get waves. Mechanical waves require a material medium for their propagation, such as water or air, while electromagnetic waves can travel through a vacuum. It is important to distinguish between longitudinal waves, where particle displacement is parallel to wave direction, and transverse waves, where displacement is perpendicular. In your exams, you must be able to identify these types and explain how energy is carried by the disturbance. Remember that the speed of a wave is determined by the properties of the medium, not the source, which is a common point of confusion for many students.

Key Points to Remember:

1. Mechanical waves need a medium to travel.
2. Longitudinal waves involve compressions and rarefactions.
3. Transverse waves involve crests and troughs.
4. The restoring force is proportional to displacement in SHM.
5. Wave speed depends on the medium properties.

2.2 Characteristics of Waves

To fully describe a wave, we must understand its specific characteristics: amplitude, wavelength, frequency, and period. The amplitude is the maximum displacement from the mean position, which relates to the energy of the wave. The wavelength is the distance between two consecutive crests or troughs, while the frequency is the number of waves passing a point per second. These variables are linked by the fundamental wave equation, v = f - lambda. In the Federal Board exams, you will often be asked to calculate one of these variables given the others. It is vital to maintain consistent units, usually SI units like meters for length and Hertz for frequency. Understanding these characteristics allows us to predict how a wave will behave when it encounters obstacles or changes medium, which is a core competency for your physics assessment.

Key Points to Remember:

1. Amplitude represents the maximum energy displacement.
2. Frequency is the inverse of the time period.
3. Wavelength is measured between identical points on a wave.
4. The wave equation v = f - lambda is universal.
5. Period is the time taken for one complete vibration.

2.3 Properties of Sound Waves

Sound is a longitudinal mechanical wave produced by vibrating bodies. As an object vibrates, it creates regions of high pressure called compressions and low pressure called rarefactions in the surrounding air. The human ear detects these pressure variations as sound. The quality of sound is determined by its frequency, which we perceive as pitch, and its amplitude, which we perceive as loudness. A high-frequency sound has a high pitch, while a large amplitude sound is loud. Furthermore, the speed of sound varies significantly depending on the temperature and density of the medium. In solids, sound travels faster than in liquids or gases because the particles are more closely packed. Being able to explain these dependencies is crucial for answering the conceptual questions that appear in the Federal Board papers every year.

Key Points to Remember:

1. Sound requires a medium to propagate.

2. Pitch is directly related to the frequency of sound.
3. Loudness is directly related to the amplitude of vibration.
4. Sound travels fastest in solids and slowest in gases.
5. Compressions and rarefactions form the sound wave.

2.4 Reflection and Refraction of Sound

Sound waves obey the same laws of reflection and refraction as light waves. Reflection of sound occurs when sound waves bounce off a surface, creating an echo if the distance is sufficient to distinguish the reflected sound from the original. This principle is used in sonar and ultrasound technology. Refraction of sound occurs when sound waves pass from one medium to another or through regions of different temperatures, causing the wave to bend. This explains why sound often travels further at night when the air near the ground is cooler. For your board exams, you should be prepared to discuss the conditions required for an echo, such as the minimum distance of 17 meters, and how these phenomena are applied in real-world scenarios like medical imaging or architectural design for concert halls.

Key Points to Remember:

1. Echo requires a minimum distance of 17 meters.
2. Reflection of sound is utilized in sonar systems.
3. Refraction causes sound to bend due to temperature changes.
4. Sound waves follow the law of reflection: angle i = angle r.
5. Reverberation is the persistence of sound due to multiple reflections.

3. Key Definitions, Formulas, and Rules

1. **Wave Equation:** v = f - lambda
2. **Frequency:** f = 1 / T
3. **Time Period:** T = 1 / f
4. **Speed of Sound:** v = d / t
5. **Simple Harmonic Motion:** F = -k - x
6. **Echo Distance:** d = (v - t) / 2
7. **Wavelength:** Distance between two consecutive crests or troughs.
8. **Amplitude:** Maximum displacement from the mean position.
9. **Longitudinal Wave:** Wave where particles oscillate parallel to the direction of wave propagation.

4. Step-by-Step Worked Examples

Example 1: Calculating Wave Speed

Problem: A wave has a frequency of 50 Hz and a wavelength of 0.4 meters. Calculate the speed of the wave.

Solution Steps:

Step 1: Identify given values: f = 50 Hz, lambda = 0.4 m.

Step 2: Use the wave equation: v = f - lambda.

Step 3: Substitute values: v = 50 - 0.4.

Step 4: Calculate the result: v = 20 m/s.

Final Answer: 20 m/s

Example 2: Finding Time Period

Problem: If a pendulum completes 20 vibrations in 10 seconds, what is its time period?

Solution Steps:

Step 1: Identify given values: Total vibrations = 20, Total time = 10 s.

Step 2: Calculate frequency: f = vibrations / time = 20 / 10 = 2 Hz.

Step 3: Use the relation T = 1 / f.

Step 4: Calculate the result: T = 1 / 2 = 0.5 s.

Final Answer: 0.5 seconds

Example 3: Echo Calculation

Problem: A person shouts towards a cliff and hears the echo after 3 seconds. If the speed of sound is 340 m/s, how far is the cliff?

Solution Steps:

Step 1: Identify given values: v = 340 m/s, t = 3 s.

Step 2: Use the echo formula: d = (v - t) / 2.

Step 3: Substitute values: d = (340 - 3) / 2.

Step 4: Calculate: d = 1020 / 2 = 510 m.

Final Answer: 510 meters

Example 4: Wavelength Determination

Problem: A radio wave travels at 3 - 10^8 m/s with a frequency of 100 MHz. Find its wavelength.

Solution Steps:

Step 1: Convert frequency to Hz: 100 MHz = 100 - 10^6 Hz = 10^8 Hz.

Step 2: Use the formula: lambda = v / f.

Step 3: Substitute values: lambda = (3 - 10^8) / 10^8.

Step 4: Calculate: lambda = 3 m.

Final Answer: 3 meters

5. Common Exam Mistakes to Avoid

1. Forgetting to convert units, such as MHz to Hz or milliseconds to seconds.
2. Confusing the formula for echo by forgetting to divide the total distance by two.
3. Mixing up the definitions of frequency and time period in numerical problems.
4. Assuming sound travels at the same speed in all media, ignoring density and temperature effects.
5. Failing to label units in the final answer, which leads to marks deduction in board exams.
6. Confusing transverse and longitudinal wave definitions during conceptual explanations.

6. Practice Questions and Model Answers

Question 1

Define longitudinal waves and give one example.

Answer 1: Longitudinal waves are waves in which the particles of the medium oscillate parallel to the direction of wave propagation. Example: Sound waves.

Question 2

Why does sound travel faster in solids than in gases?

Answer 2: Sound travels faster in solids because the particles are more closely packed and have stronger intermolecular forces, allowing for faster energy transfer between them compared to the sparse particles in gases.

Question 3

Calculate the frequency of a wave with a period of 0.02 seconds.

Answer 3: Using f = 1 / T, f = 1 / 0.02 = 50 Hz.

Question 4

What is the difference between pitch and loudness?

Answer 4: Pitch is the characteristic of sound that depends on frequency (higher frequency equals higher pitch), while loudness depends on amplitude (larger amplitude equals higher loudness).

Question 5

Can sound travel through a vacuum? Explain.

Answer 5: No, sound cannot travel through a vacuum because it is a mechanical wave that requires a material medium (particles) to transfer energy from one point to another.